

SIMATS ENGINEERING

ASSESSMENT TOOL 1: Concept Mapping

COURSE NAME: Cloud Computing and
Big Data Analytics

COURSE CODE: CSA15

COURSE OUTCOME COVERED

CO1: *Apply cloud computing technologies including hardware-software evolution, virtualization, web services, and service models to develop cloud-based solutions. (BL2)*

Assessment Tool Weightage: 20%

Dave's Taxonomy: Imitation (Level 1)
Question Count: 4

Total Marks: 50

Code of Conduct

I **Yarra Pawan (192424294)** certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: *Y Pawan*

Assessment Tasks

Concept Mapping Tasks

Create concept maps for the following tasks. Each map must show key nodes, link labels, and a logical hierarchy.

1. Concept Map 1: Cloud Computing Fundamentals. Include definition, characteristics, benefits, and limitations.
2. Concept Map 2: Evolution of Cloud Computing. Connect hardware evolution, internet software evolution, distributed systems, and service delivery.
3. Concept Map 3: Service Models. Map IaaS, PaaS, SaaS, and XaaS with provider responsibilities and user responsibilities.
4. Concept Map 4: Virtualization and Web Services as Enablers of Cloud. Show how these two concepts support cloud deployment and scalability.

Expected concept nodes:

- Elasticity, scalability, resource pooling, on-demand access, measured service
- Virtual machine, hypervisor, abstraction, API, SOAP, REST, provider, consumer
- Infrastructure layer, platform layer, application layer, shared responsibility

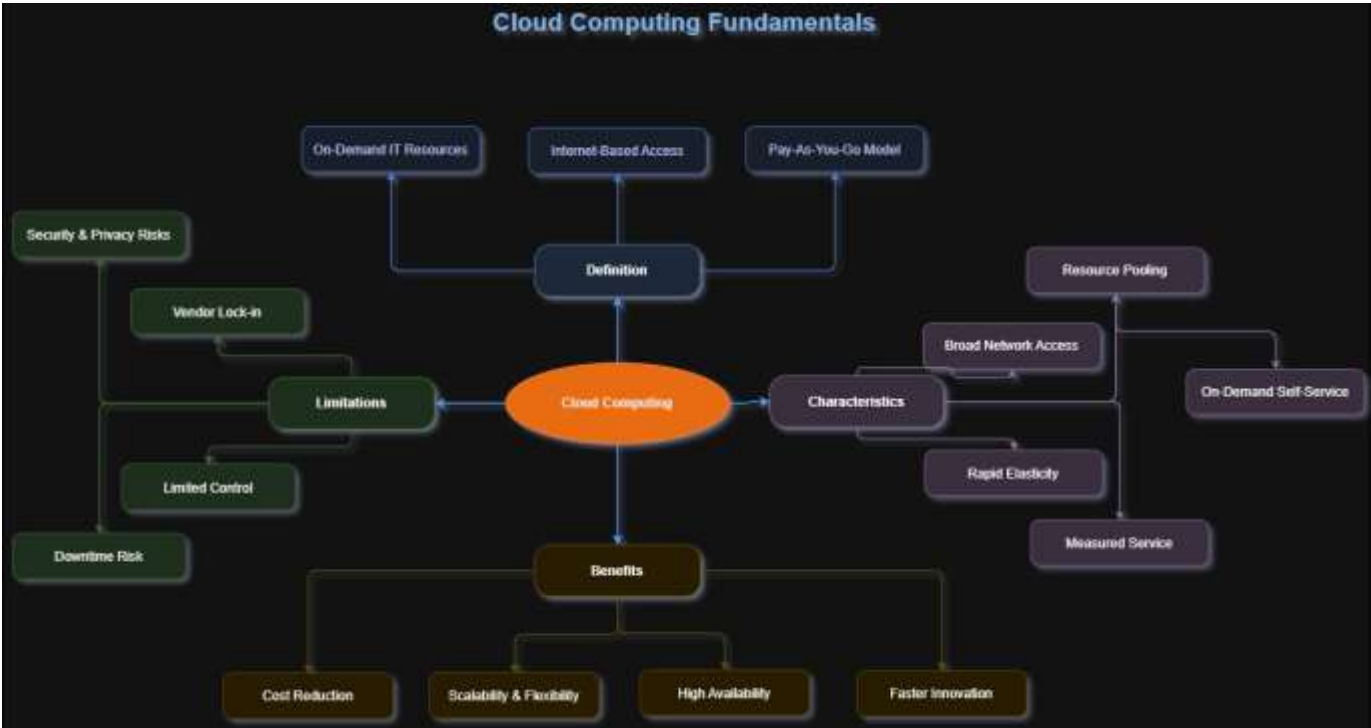
Concept Mapping Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Concept Identification	25%	All major concepts are accurately identified	Most concepts are identified correctly	Limited concepts are identified	Essential concepts are missing
Relationship Mapping	25%	Links between concepts are	Most links are correct	Some links are weak or unclear	Relationships are mostly incorrect

		accurate and meaningful			
Hierarchy and Organization	20%	Excellent structure with clear hierarchy	Mostly organized with minor issues	Basic structure with weak hierarchy	No logical organization
Technical Relevance	20%	Map strongly reflects cloud course content	Mostly aligned to topic	Partially aligned to topic	Weak alignment to topic
Visual Clarity	10%	Neat, readable, and easy to interpret	Generally readable	Somewhat cluttered	Poorly presented

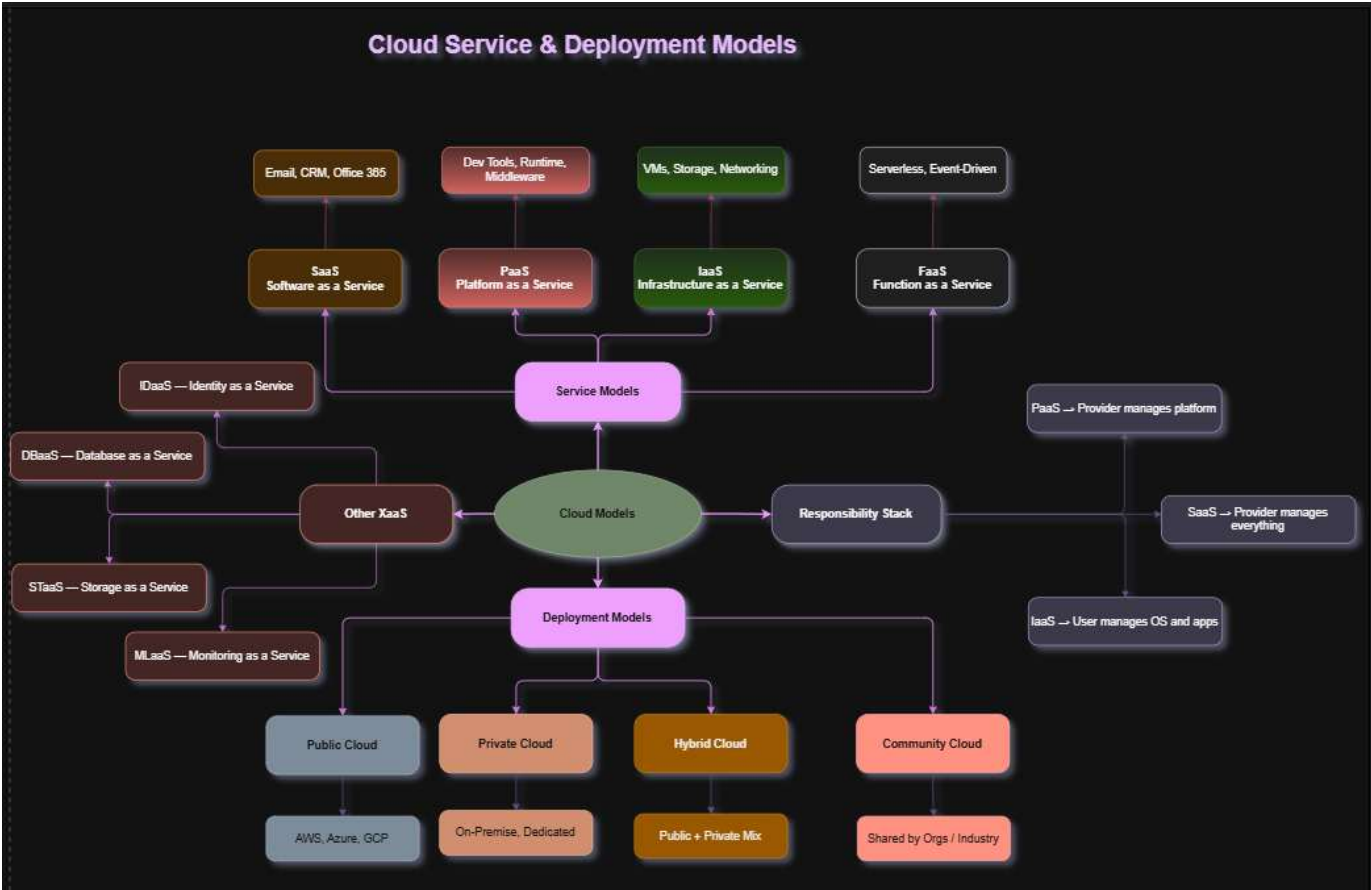
Concept Map 1: Cloud Computing Fundamentals. Include definition, characteristics, benefits, and limitations.

Tools Used: draw.io



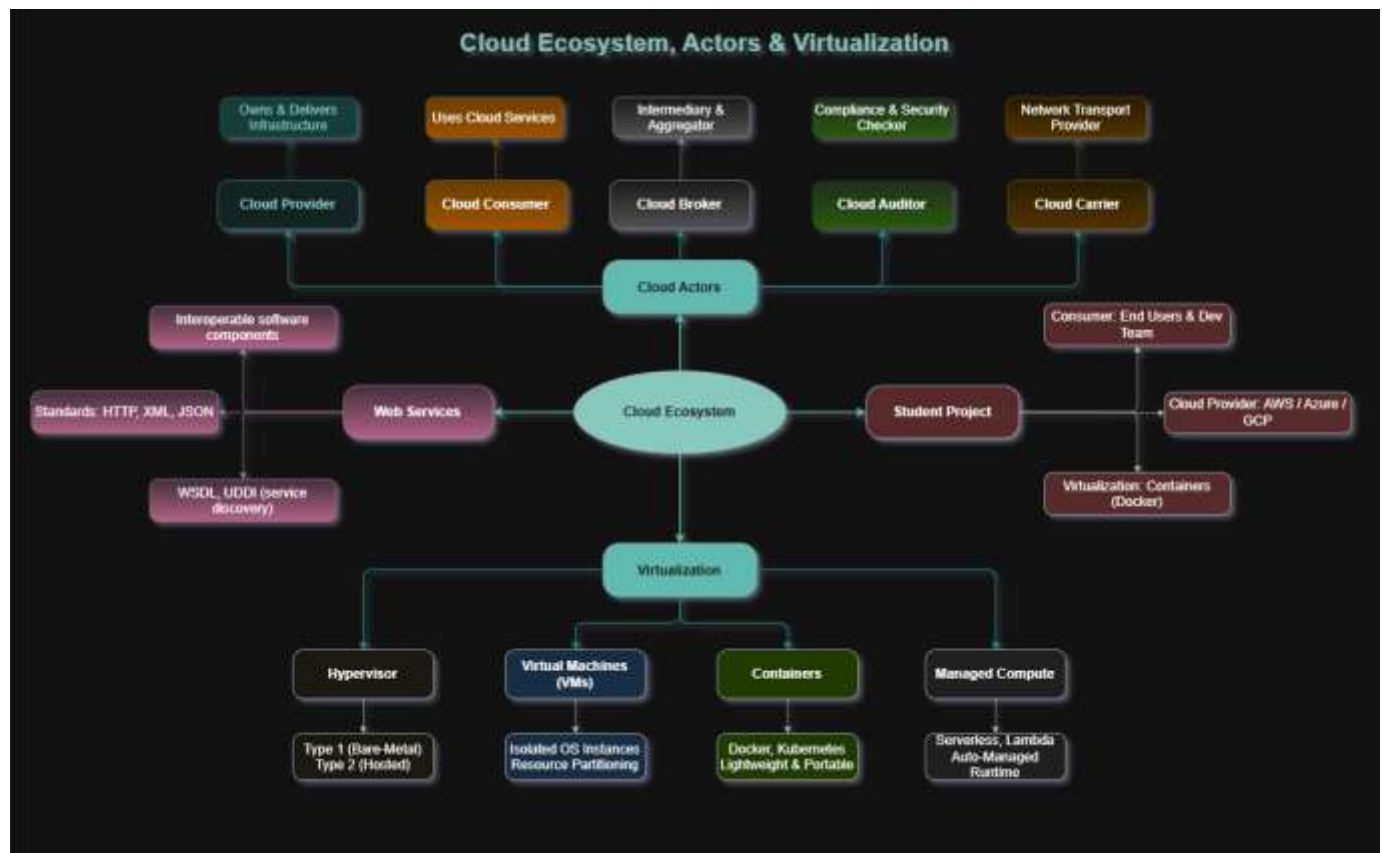
Concept Map 2: Evolution of Cloud Computing. Connect hardware evolution, internet software evolution, distributed systems, and service delivery.

Tools Used: draw.io



Concept Map 3: Service Models. Map IaaS, PaaS, SaaS, and XaaS with provider responsibilities and user responsibilities.

Tools Used: draw.io



Concept Map 4: Virtualization and Web Services as Enablers of Cloud. Show how these two concepts support cloud deployment and scalability.

Tools Used: draw.io



1. Evaluate the effectiveness of your concept map in representing cloud computing principles. Which design decisions most strongly support your solution, and why?
2. Analyze how scalability, elasticity, and resource management influence the architecture shown in your concept map. What challenges could arise if these capabilities were absent?
3. Justify the selection of the APIs, web services, or cloud services included in your concept map. How do they contribute to the overall functionality and efficiency of the system?
4. Critically examine the relationships between the components in your concept map. How do these interactions improve reliability, performance, or user experience?
5. Reflect on the process of developing your concept map. How has creating and analyzing the map changed your understanding of cloud computing architecture, and what improvements would you make if you were redesigning it?

Task 4:

1. Evaluate the effectiveness of your concept map in representing cloud computing principles. Which design decisions most strongly support your solution, and why?

The concept map effectively represents the fundamental principles of cloud computing by organizing the major concepts into a logical and hierarchical structure. The central node, Cloud Computing Ecosystem, acts as the foundation from which all related components branch out, allowing the viewer to understand how different cloud technologies are interconnected. The map includes cloud actors, service models, deployment models, virtualization, web services, REST/SOAP APIs, scalability, elasticity, and the project architecture, ensuring comprehensive coverage of the concepts required in CO1.

One of the strongest design decisions is the use of grouped categories, where related concepts are placed together. For example, service models (IaaS, PaaS, SaaS, and XaaS) are grouped separately from deployment models, making it easier to distinguish between service delivery and deployment strategies. Similarly, virtualization and web services are presented as enabling technologies that support cloud infrastructure and communication.

Another important design choice is the use of labeled connectors such as *provides*, *enables*, *communicates through*, *stores data in*, and *scales with*. These relationship labels clearly explain how each component interacts with others instead of simply connecting nodes with arrows. Color coding, icons, and consistent formatting further improve readability and help users quickly identify different sections of the cloud ecosystem.

Overall, the concept map successfully demonstrates cloud computing principles in a visual and structured manner. It not only explains theoretical concepts but also relates them to a practical cloud-based project, making the learning process more meaningful and easier to understand.

2. Analyze how scalability, elasticity, and resource management influence the architecture shown in your concept map. What challenges could arise if these capabilities were absent?

Scalability, elasticity, and resource management are three of the most important characteristics of cloud computing, and they significantly influence the architecture represented in the concept map. Scalability enables the cloud system to handle increasing numbers of users, transactions, or workloads by adding additional computing resources. This can be achieved through vertical scaling, where hardware resources such as CPU and memory are increased, or horizontal scaling, where additional servers or virtual machines are added to distribute the workload.

Elasticity complements scalability by automatically increasing or decreasing resources based on current demand. During peak usage, such as a large number of ride requests in the student's project, cloud resources can automatically expand to maintain system performance. When demand decreases, unnecessary resources are released, reducing operational costs and improving resource efficiency.

Resource management ensures that processing power, storage, memory, and networking resources are allocated efficiently among multiple users and applications. Technologies such as virtualization, load balancing, and monitoring help optimize resource utilization while maintaining reliability and availability.

If these capabilities were absent, several problems could occur. High user traffic could overload servers, causing slow response times, application crashes, or service interruptions. Organizations would need to purchase additional hardware even when it is not continuously required, increasing infrastructure costs. Poor resource allocation could also lead to wasted computing resources, reduced performance, and lower customer satisfaction. Therefore, scalability, elasticity, and resource management are essential for building reliable, cost-effective, and high-performance cloud applications.

3. Justify the selection of the APIs, web services, or cloud services included in your concept map. How do they contribute to the overall functionality and efficiency of the system?

The concept map includes several APIs, web services, and cloud services because each one plays an important role in enabling communication, security, storage, scalability, and application management. REST APIs are selected as the primary communication mechanism because they are lightweight, stateless,

easy to integrate, and widely supported by modern web and mobile applications. REST uses standard HTTP methods such as GET, POST, PUT, and DELETE, making communication between the client application and backend server simple and efficient.

SOAP APIs are also included to demonstrate support for enterprise applications that require standardized messaging, enhanced security, and reliable transactions. While REST is more commonly used in modern cloud applications, SOAP remains important for financial systems, government applications, and enterprise integrations.

Virtualization technologies enable multiple virtual machines to share the same physical hardware, improving resource utilization and reducing infrastructure costs. Cloud services such as MySQL Database, Railway cloud hosting, Google Maps API, Firebase Authentication, and notification services support different aspects of the project. The database provides reliable data storage, authentication services protect user identities, and notification services enable real-time communication with users.

These services work together to improve application performance, security, scalability, and availability. APIs simplify communication between system components, cloud services reduce infrastructure management overhead, and managed platforms allow developers to focus on application development instead of maintaining servers. Together, these technologies create a reliable, flexible, and efficient cloud-based solution.

4. Critically examine the relationships between the components in your concept map. How do these interactions improve reliability, performance, or user experience?

The relationships between the components in the concept map illustrate how different cloud technologies cooperate to deliver efficient and reliable cloud services. The cloud actors initiate requests through the user application, which communicates with the backend using REST APIs. The backend processes requests, authenticates users, performs business logic, and stores or retrieves information from the cloud database. This layered architecture separates responsibilities among different components, making the system easier to maintain and expand.

Virtualization forms the foundation of cloud infrastructure by allowing multiple virtual machines to share physical resources efficiently. This enables cloud providers to allocate resources dynamically based on application demand. Service models such as IaaS, PaaS, and SaaS define different levels of responsibility between the cloud provider and the application developer, ensuring efficient resource management and simplified deployment.

Scalability and elasticity work together to improve performance by automatically adjusting computing resources according to workload. Load balancing distributes incoming requests across multiple servers, preventing bottlenecks and improving system reliability. Monitoring and logging continuously observe system performance, allowing administrators to detect and resolve issues before they affect users.

These interactions significantly enhance the user experience by providing faster response times, higher system availability, secure authentication, and uninterrupted access to services. Even during periods of heavy traffic, the cloud infrastructure maintains stable performance through automatic resource allocation. Overall, the relationships between these components create a flexible, secure, and highly reliable cloud computing environment.

5. Reflect on the process of developing your concept map. How has creating and analyzing the map changed your understanding of cloud computing architecture, and what improvements would you make if you were redesigning it?

Developing the concept map was a valuable learning experience because it required me to analyze how different cloud computing concepts are connected rather than studying them individually. While creating the map, I gained a deeper understanding of how cloud actors, deployment models, service models, virtualization, APIs, scalability, elasticity, and cloud infrastructure interact to build complete cloud-based applications. Organizing these concepts visually made it easier to understand the overall architecture and the flow of information between different cloud components.

The process also improved my understanding of the shared responsibility model in cloud computing. I learned how cloud providers manage infrastructure, virtualization, and platform services, while developers focus on application development, business logic, and data management. Relating these concepts to the Slipstream ride-sharing project helped me understand how cloud technologies are applied in real-world software systems rather than remaining theoretical concepts.

If I were redesigning the concept map, I would include additional modern cloud technologies such as Docker containers, Kubernetes orchestration, microservices architecture, serverless computing, DevOps pipelines, and AI-based cloud services. I would also improve the visual presentation by adding more cloud architecture icons, interactive layers, and clearer relationship labels to represent data flow and service interactions. These improvements would make the concept map even more comprehensive, technically accurate, and easier to understand for both students and evaluators.

Conclusion:

The concept map successfully represents the core principles of cloud computing by connecting cloud actors, service models, deployment models, virtualization, web services, APIs, scalability, elasticity, and the project architecture into a single, well-organized framework. It demonstrates how these components work together to deliver secure, reliable, scalable, and cost-effective cloud solutions. Through this activity, I gained a better understanding of the relationships between cloud technologies and their practical application in developing modern software systems. The exercise also highlighted the importance of virtualization, resource management, and cloud service models in achieving high performance and availability. Overall, the concept map effectively satisfies the Course Outcome (CO1) by illustrating both the theoretical concepts and their real-world implementation in a cloud-based project.